

8. The electromagnetic (em) spectrum consists of seven regions.
Two of these regions are infra-red and microwaves.
Some information about these regions is given in the table below.

Property	Infra-red	Microwaves
Wave speed in space (m/s)		
Typical wavelength (m)	7.5×10^{-7}	
Typical frequency (Hz)	4.0×10^{14}	3.0×10^{10}

- (a) Use an equation from page 2 to **complete the table**. [4]
Space for working.

- (b) Communication between the UK and USA can be made by optical fibres or by geostationary satellites.
An infra-red signal sent by optical fibre travels 9000 km and a microwave signal sent by satellite travels 72 000 km.

The speed of the microwave signal through space is 1.5 times the speed of the infra-red signal through glass.

- (i) Compare these methods in terms of time delay between sending a signal from the UK and receiving the signal in the USA. [2]
Space for working.

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- (ii) Suggest **two** advantages of sending signals by optical fibre from the UK to the USA. [2]

1.
2.



- (iii) A geostationary satellite stays directly above the same point on the Earth.
It orbits in the same direction as the Earth spins.

State the **two** other conditions needed to achieve geostationary orbit.

[2]

1.

2.

Examiner
only

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